

What's the default size of stack and heap, and what are the considerations?

- **Stack (default size):**
 - Typically **1 MB per thread** on Windows for .NET applications (can vary by OS, architecture, and runtime).
- **Heap (default size):**
 - No fixed size; it **grows dynamically** up to the limits of the system's available memory.

Considerations:

- **Stack** is faster but limited in size. Used for small, short-lived variables and function calls. Exceeding its limit causes a **StackOverflowException**.
 - **Heap** is larger and used for dynamic memory (e.g., objects). It's managed by the **garbage collector**, but access is slower compared to the stack.
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What is time complexity?

Time complexity is a way to describe how the **execution time of an algorithm** changes with the size of its input.

- Expressed using **Big O notation** (e.g., $O(1)$, $O(n)$, $O(\log n)$, $O(n^2)$)
- Helps estimate **efficiency** and **scalability**

Example:

- $O(1)$ = constant time
- $O(n)$ = time grows linearly with input size
- $O(n^2)$ = time grows quadratically